Pictorial Essay

An Atlas of Gallium-68-fibroblast Activation Protein Inhibitor (FAPI) in Oncology and Nononcology Imaging

Abstract

Fibroblast activation protein (FAP) is a quinoline-based membrane bound glycoprotein enzyme. It is not expressed in normal adult tissue except for the myometrium and sometimes in pancreas. The expression increases in inflammation and in cancer-associated fibroblasts. Gallium 68-FAP inhibitor (FAPI) is known to localize in various neoplasms by the virtue of their property of cancer-associated fibroblast activity. Fibroblasts are also known to be associated in inflammatory and infectious condition such as cholecystitis, pyelonephritis, and tuberculosis. Immunoglobulin G4 related disease (IgG4RD) is characterized by infiltration of various tissues with IgG4-rich lymphoplasmacytic fibrotic tissue and has been reported to express cancer-associated fibroblast. The time-tested molecule for cancer imaging F18-fluorode-oxyglucose requires 4–6 h of fasting state. Ga68-FAPI does not require fasting. This pictorial essay illustrates various entities, cancerous as well as noncancerous using Ga68-FAPI positron emission tomography computed tomography (PET CT) imaging.

Keywords: *Gallium68-fibroblast activation protein inhibitor, nononcology, oncology*

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Introduction

Gallium-68 fibroblast activation protein inhibitor (FAPI) is a novel compound that specifically targets cancer-associated fibroblasts (CAFs). FAP, a quinoline-based radiotracer, is expressed in dimeric form on the surface of CAFs. These fibroblasts play a crucial role in promoting tumor growth, angiogenesis, and are directly associated with adverse prognosis. CAFs are derived from various cell types, including mesenchymal cells, fibrocytes from bone marrow, and cells undergoing mesenchymal transition.[1] In addition, they interact with cancer cells and contribute to extracellular matrix deposition. Notably, CAFs exhibit an elongated morphology and lack endothelial, epithelial, and leukocyte markers.[2]

In addition to being expressed on CAFs, FAP is also overexpressed on tumor-associated macrophages, which play a role in immunesuppression. CAFs can constitute up to 80%–90% of the tumor volume.[3] When tumors grow larger than 2 mm, CAFs become responsible for supplying nutrients to the tumor cells.[4] Unlike normal fibroblasts, those associated with cancer express FAP and secrete higher levels of extracellular proteins, promoting tumor cell proliferation and growth.[5] Normal fibroblasts have a spindle-shaped morphology, while CAFs exhibit a cruciform or stellate shape.[6] Fibroblast-associated protein inhibitors were initially developed as anticancer agents and have since been explored for theranostics.[7] Indeed, FAPα, which is targeted by FAPI, exhibits a fascinating expression pattern. FAP alpha is also known as prolyl endopeptidase FAP which modulates tissue remodelling, tumor growth, invasion, and immunosuppression. Further details are as follows:

1. FAPα is expressed at high levels in several types of tumors, making it a promising target for imaging and therapy
2. Among various tumor entities, solitary fibrous tumors (SFTs) have particularly high FAPα expression. SFTs are rare mesenchymal neoplasms
3. FAPI positron emission tomography/computed tomography (PET/CT) imaging has demonstrated increased tumor uptake in patients with SFTs, providing an improved detection efficacy compared to fluorodeoxyglucose (FDG) PET
4. FAPα is primarily located on the tumor cell surface in SFTs.

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Expression beyond tumors: Interestingly, FAPα is not exclusive to tumors. It is also overexpressed in benign conditions, such as wound healing, fibrosis, arthritis, and atherosclerotic plaques. Inflammatory and immune-mediated pathologies may also exhibit FAPα expression. FAPI imaging holds potential for distinguishing between benign and malignant lesions in given clinical contexts.

Clinical Applications

Fibroblast activation protein inhibitor (FAPI) PET/CT can detect a wide range of malignancies, including head-and-neck cancers, lung cancers, colorectal cancers, esophageal cancers, and breast cancers. It has been shown to detect 28 different types of malignancies in one study.[8]

Notably, FAPI shows low background uptake in the liver, making it useful for detecting liver primary tumors and metastases.[9]

In summary, FAPIs versatility extends beyond tumors, offering exciting possibilities for both diagnostic imaging and potential therapeutic applications. Further research will define its role in patient care.

Method

Radiopharmacy

Ge 68-Ga 68 generator (iTG: iQS Ga 68 Fluiding Labeling Module-semi automatic method) was used to obtain Ga 68 FAPI (The cold kit/carrier molecule being FAPI 46).

The cold kit was dispensed in Na acetate buffer (1.25 g Sodium acetate in 50 mL high-performance liquid chromatography water (HPLC) water) and preheated to 1050 C, as the reaction temperature.

The eluent used was 4 mL of 0.05M HCl. The reaction time was 10 min.

The C-8 Cartridge was used to purify the product after its preconditioning with 70% ethanol (35 mL ethanol + 15 mL HPLC water).

The product was then diluted with 5 mL saline and 5 mL 60% ethanol before IV administration to the patient, through the secured IV line.

The administered dose was 3–4 mCi per case, based on the weight of the patient (0.05–0.06 mCi/kg of body weight).

Scintigraphy

As Gallium-68-FAPI is a new molecule, an ethics committee approval was obtained. Informed consent from all patients undergoing the test was procured. Whole-body imaging was acquired between 50- and 60-min postradiotracer administration using Siemens Horizon 16 slice PET CT system. Images were acquired at the rate of 120 s per bed. Intravenous contrast was not used. Delayed scan after diuretic administration was acquired for the evaluation of pyelonephritis. Images were reconstructed in short axis, sagittal, and coronal planes by Fourier rebinning (Fore) + filtered back projection method. Scans were read by one specialist and validated by second.

Discussion

- Fibroblast activation plays a crucial role in tissue repair and fibrosis; here are some key points: Fibroblast activation: Fibroblasts are mesenchymal cells involved in maintaining tissue integrity and acting as local progenitor reservoirs.[10] In response to tissue injury, fibroblasts can transition into myofibroblasts, characterized by extracellular matrix secretion and actomyosin-based stress fiber contraction.[11] Transient activation into myofibroblasts aids tissue repair, while persistent activation leads to pathological fibrosis[12,13]
- Fibroblast behavior in tumor: Fibroblasts exhibit dynamic behavior during tumor progression. The process of fibroblast evolution in tumor cells is termed "stromagenesis." Initially, fibroblasts may restrict or suppress tumor growth, but subsequent microenvironment changes promote tumor proliferation[14]
- FAP: Fibroblast activation protein is associated with tumor development and poor clinical outcomes in various cancers.[15,16] It localizes not only in malignant conditions but also in benign ones such as inflammatory arthritis, vasculitis, and cirrhosis.[17,18] Ga-68-FAPI imaging demonstrated utility across different entities as follows.

Lymphoma

Cancer associated fibroblasts (CAFs) overexpress FAP, which is enriched in many epithelial carcinomas and hematologic neoplasms. Ga-68-FAPI PET/CT is a novel diagnostic tool that visualizes the tumor stroma by targeting FAP. A prospective study evaluated lymphoma patients using Ga-68-FAPI PET/CT and correlated FAP uptake with immunostaining intensity. Hodgkin lymphoma (HL) lesions showed significantly elevated FAP uptake, correlating with intense FAP immunostaining (score, 3+). Aggressive non-HL (NHL) lesions (moderate-to-strong FAP staining) exhibited intense to moderate Ga-68-FAPI uptake. Indolent NHL lesions showed weak FAP staining and mild-to-moderate Ga68-FAPI uptake.[19] Ga-68-FAPI imaging can detect FAP expression in lymphoma lesions. It may serve as an alternate method for characterizing lymphoma profiles. Notably, Ga-68-FAPI PET/CT provides valuable information beyond traditional FDG imaging, especially in cases such as liver lymphoma with lower physiological FDG uptake.[20] Figure 1 illustrates FAPI avid multiple infra-diaphragmatic nodes.

Pyelonephritis

1. Acute pyelonephritis: Acute pyelonephritis is a bacterial infection of the kidney and renal pelvis. It should be

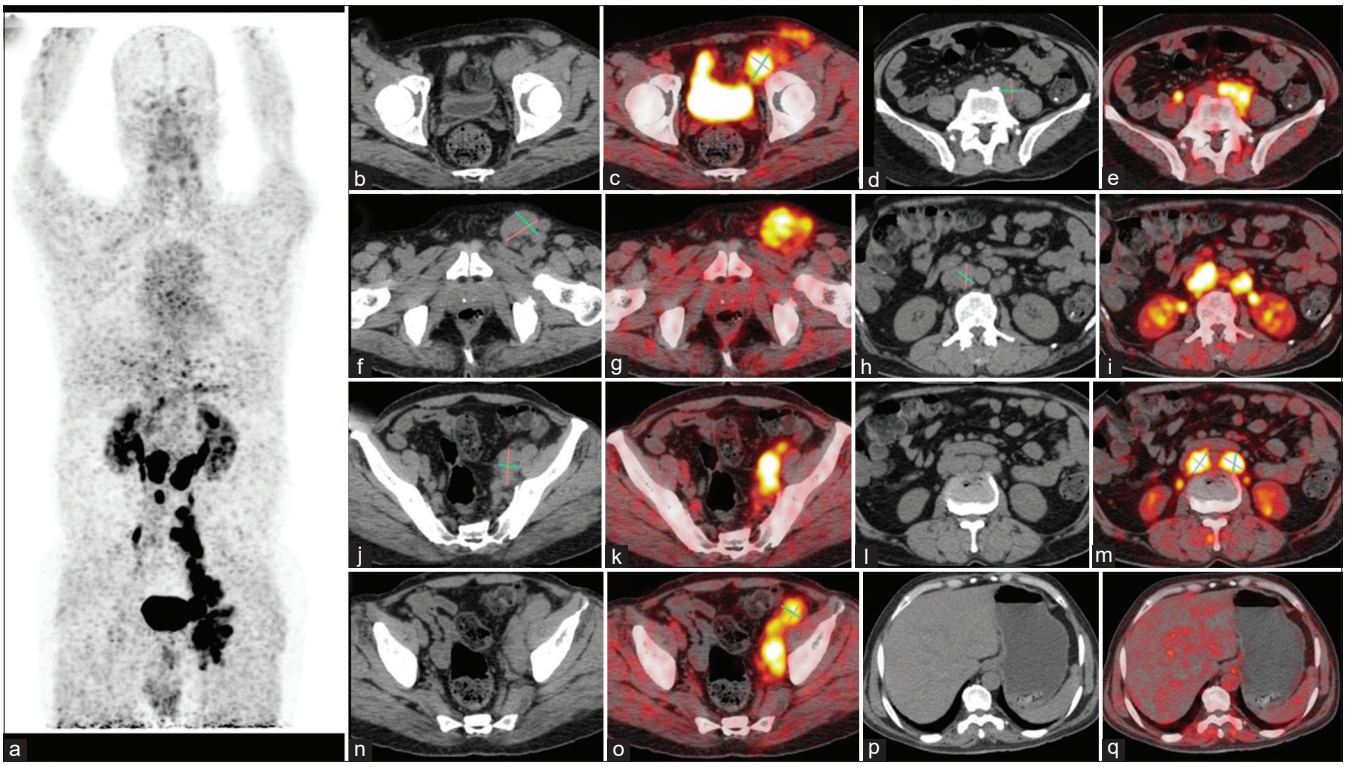


Figure 1: A 62-year-old male had 7 kg weight loss, anorexia, general weakness, and itching over the body. General examination was unremarkable. Blood test showed mild anaemia. Fibroblast activation protein inhibitor positron emission tomography computed tomography scan showed increased tracer uptake at multiple sites in abdomen and pelvis on maximum intensity projection (MIP) images (a). FAPI uptake in enlarged abdominal (h, i, l and m) - Pelvic (b-e, j, k, n and o) and left inguinal (f and g) lymph nodes. Spleen was not enlarged and there were no focal splenic lesions (p and q). Histology was non-Hodgkin lymphoma

suspected in patients with pyrexia, flank pain, and laboratory evidence of urinary tract infection. Urine culture with antimicrobial susceptibility testing guides antibiotic therapy

2. Imaging in pyelonephritis: Contrast-enhanced CT is the preferred modality for evaluating acute bacterial pyelonephritis. CT detects focal parenchymal abnormalities, defines disease extent, and identifies perinephric fluid collections or abscesses. When contrast or radiation is contraindicated (e. g., during pregnancy), ultrasonography and magnetic resonance imaging (MRI) are alternative options
3. FAPI imaging: FAPI imaging demonstrated intense tracer uptake in the bilateral renal cortex. Renal pelvis does not show tracer uptake in view of postdiuretic delayed scan. MRI revealed altered renal cortical signals. Dimercaptosuccinic acid renal scan confirmed bilateral renal cortical cold areas consistent with pyelonephritis [Figure 2].[21]

Ga-68-fibroblast Activation Protein Inhibitor Positron Emission Tomography/Computed Tomography in Ovarian Neoplasms

FAPI imaging demonstrates intense tracer uptake in bilateral adnexa (ovarian regions) in ovarian neoplasms. FAP expression is found in more than 90% of epithelial ovarian cancers, including serous and mucous subtypes. Notably, FAP is absent or low in normal ovarian tissue and benign conditions. Dendl K *et al*. reported variable FDG uptake in ovaries based on the stage of ovulation. In contrast, FAPI PET does not exhibit such variability, suggesting it's promising role in ovarian neoplasm.[22] Our case demonstrates avid uptake of FAPI in bilateral adnexa in ovarian neoplasm with synchronous lesion in the left breast [Figure 3].

In summary, Ga-68-FAPI PET/CT holds potential for accurate imaging in ovarian cancer.

Renal Cell Carcinoma

Renal cell carcinoma (RCC) is the most common type of kidney cancer, arising from renal tubular epithelial cells. It accounts for approximately 90% of all kidney malignancies. The common subtypes include clear cell RCC, papillary RCC, and chromophobe RCC. FAPI imaging has shown promise in detecting RCC lesions. FAP expression is elevated in RCC tissue, making it a potential target for imaging. FAPI PET/CT can provide valuable information about tumor extent and guide treatment decisions. Early-stage RCC is often asymptomatic, but as the tumor grows, symptoms may include flank pain, hematuria, and palpable abdominal masses. Surgical resection remains the primary treatment for localized RCC. In advanced cases, targeted therapies (e.g., tyrosine kinase inhibitors) and immunotherapies are used. Accurate diagnosis and staging are crucial for managing RCC

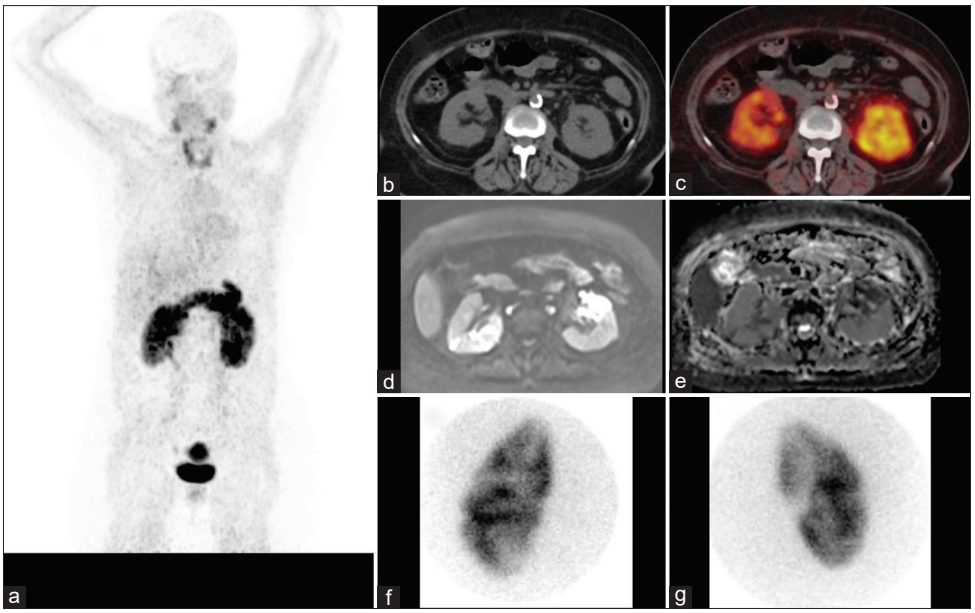


Figure 2: A 57-year-old woman with long-standing uncontrolled diabetes presented with pain in the abdomen and recurrent vomiting. General and systemic examination was unremarkable. Her blood sugar was above 500 mg/dL and hence an fluorodeoxyglucose positron emission tomography computed tomography was not possible. Ga68-fibroblast activation protein inhibitor scan showed diffuse increased tracer uptake in both the kidneys (a-c). Magnetic resonance imaging shows altered renal parenchymal signals with features of edema (d and e). Dimercaptosuccinic acid renal scan done on the next day showed bilateral renal cortical cold defects (f and g). Overall picture was compatible with the diagnosis of acute pyelonephritis

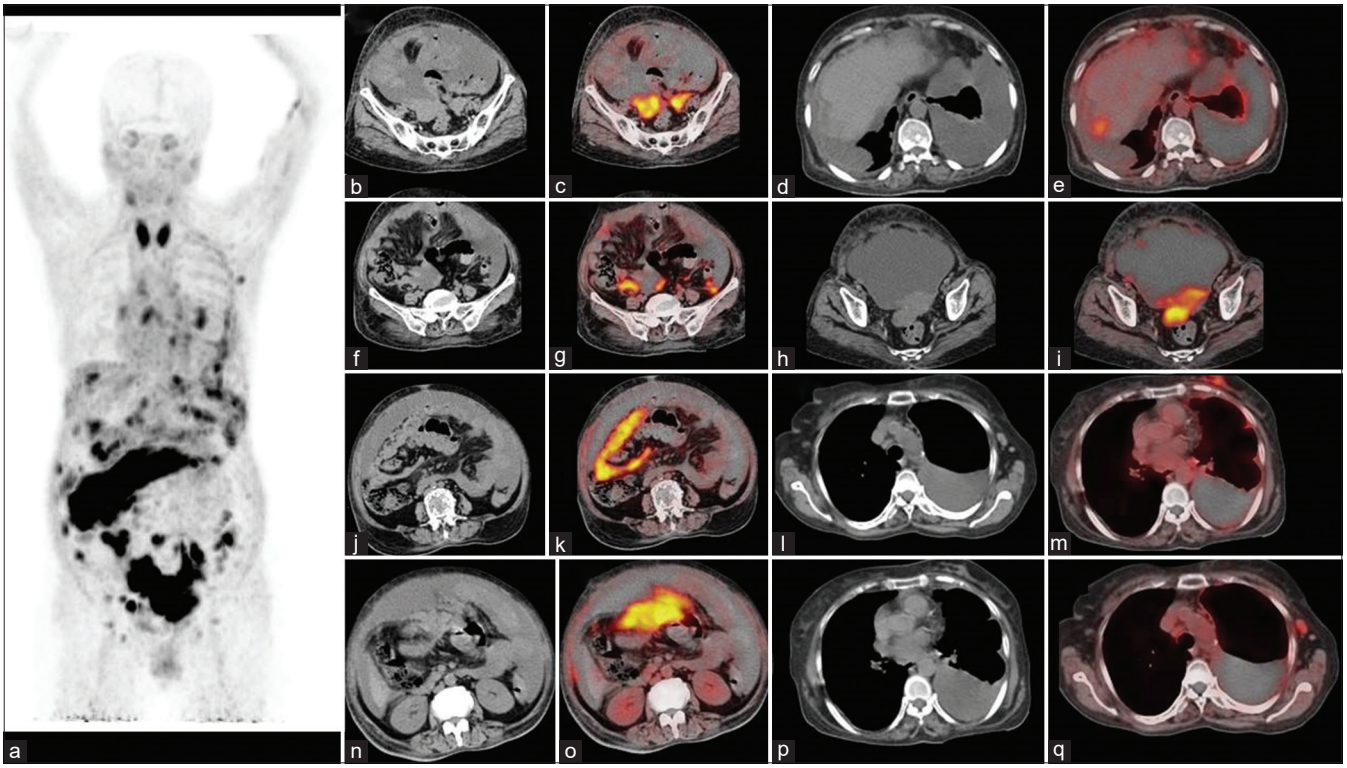


Figure 3: This 72-year-old woman presented with breathlessness, abdominal distension, and anorexia. Her ascitic fluid examination revealed mucinous adenocarcinoma cells. Fibroblast activation protein inhibitor scan showed increased tracer uptake in bilateral adnexae (b and c), pelvic peritoneal thickening-deposits (f and g), perihepatic peritoneal deposits (d and e), pouch of Douglas deposit (h and i), omental nodularity (j, k, n and o). Incidental nodule in left breast with avid tracer uptake (l and m) with ipsilateral axillary node (p and q) suggestive of synchronous primary. Maximum intensity projection (MIP) scan showed abnormal tracer localization at these sites(a). Histology was confirmatory

effectively. Figure 4 demonstrates pathological fracture in the neck of femur with few FAPI avid rib deposits. There was a suspicious FAPI avid lesion in the left kidney with corresponding cortical structural alteration on CT. This was the expected primary; however, as the patient succumbed to cardiac arrest, histologic confirmation could not be done. Dong *et al*. published a case report of false-positive FAPI in angiomyolipoma of the kidney.[23]

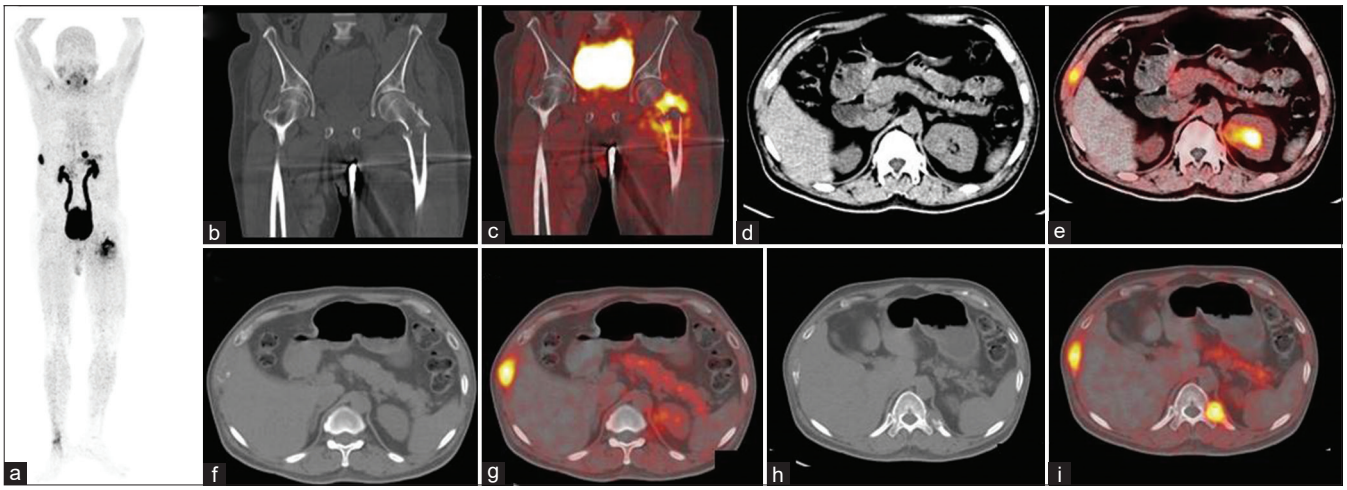


Figure 4: A 62 years old man presented with pain in hip due to pathological fracture. Fibroblast activation protein inhibitor (FAPI) scan shows abnormal tracer uptake at the fracture site (b, c), right lower rib laterally (f, g), left lower costovertebral junction (h, i). The left kidney shows a nodular lesion in superomedial cortex with avid FAPI uptake suspicious for occult primary (d, e). MIP (maximum intensity projection) image shows foci of increased tracer uptake at these sites. (a) The patient had cardiac arrest prior to biopsy

Pancreatic Ductal Adenocarcinoma

Fibroblast activation protein and pancreatic ductal adenocarcinoma

FAP is highly expressed in various tumor tissues, including pancreatic ductal adenocarcinoma (PDAC). Activated pancreatic myofibroblasts (PMFs) play a crucial role in PDAC by depositing extracellular matrix and promoting fibrosis. Key signaling events drive PMF activation, including signals from transforming growth factor-beta 1, platelet-derived growth factor, hypoxia, and extracellular matrix rigidity. FAP-expressing cells co-evolve with PDAC cells, influencing tumor proliferation by paracrine signaling. FAP expression in PDAC correlates with poor prognosis. High-FAP levels are associated with reduced overall survival and recurrence-free survival. FAP could serve as an independent prognostic factor for early-stage PDAC. Nomogram systems integrating FAP expression improve prediction of patient outcomes. Histologically, PDAC is characterized by adenocarcinoma. IHC studies reveal up to 90% FAP expression in PDAC.[24] FAP expression is variable in normal pancreas but neoplastic lesions usually show differential increased uptake.

Figure 5 illustrates a patient who presented with pain in abdomen and weight loss. FAPI showed avid uptake in the tail of pancreas with abdominal nodes. MR showed restricted diffusion in the tail of pancreas. Histology confirmed the diagnosis of adenocarcinoma of pancreas.

In summary, understanding FAP's role in PDAC biology offers potential therapeutic targets.

Oral Squamous Cell Carcinoma

Oral squamous cell carcinoma (OSCC) is the most common malignant lesion affecting the oral cavity. It arises from the thin cells lining the inner buccal mucosa and other oral structures. Early detection and accurate staging are crucial for optimal management.

FDG PET/CT versus FAP-directed PET/CT: FDG PET/CT is commonly used for OSCC imaging but may show false positives due to uptake in inflammatory cells. FAP-directed PET/CT, using a probe targeting FAP α, offers an alternative. FAP expression is elevated in cancer-associated fibroblasts, making it a potential marker for OSCC. In a study comparing F-18-FDG and FAP-directed PET/CT, both methods detected primary tumors and cervical lymph node metastases. FAP-directed PET/CT demonstrated comparable sensitivity and specificity to F-18-FDG PET/CT. Histologically, FAP expression in cancer-associated fibroblasts confirmed the findings.[25] Figure 6 shows avid FAPI uptake in squamous cell carcinoma of buccal mucosa. As muscles of oral cavity show physiologic uptake on FDG under variable conditions and FAPI diminishes the possibility of false-positive results. It has been shown to have high target to background activity by one group in 14 patients.[26]

Carcinoma of Lung

Lung carcinoma, including both non-small cell lung cancer (NSCLC) and SCLC, exhibits remarkably high uptake on Ga-68-FAPI PET/CT. The highest average SUV max (standardized uptake value) (>12) is observed in lung carcinoma, sarcoma, cholangiocarcinoma, and carcinoma of the breast. Ga-68FAPI PET/CT provides valuable insights into tumor biology and may guide clinical decisions. Our illustrated case [Figure 7] presented with malignant pleural effusion and FAPI avid mass in right lung upper lobe, regional nodes, and extensive skeletal metastases.

Wang *et al*. studied 34 patients with FAPI and FDG and observed that FAPI outperformed FDG in detection of brain, lymph node, bone, and pleural deposits.[27]

Carcinoma of Breast

FAPI imaging demonstrated intense tracer uptake in the left breast mass [Figure 8]. This uptake suggests increased

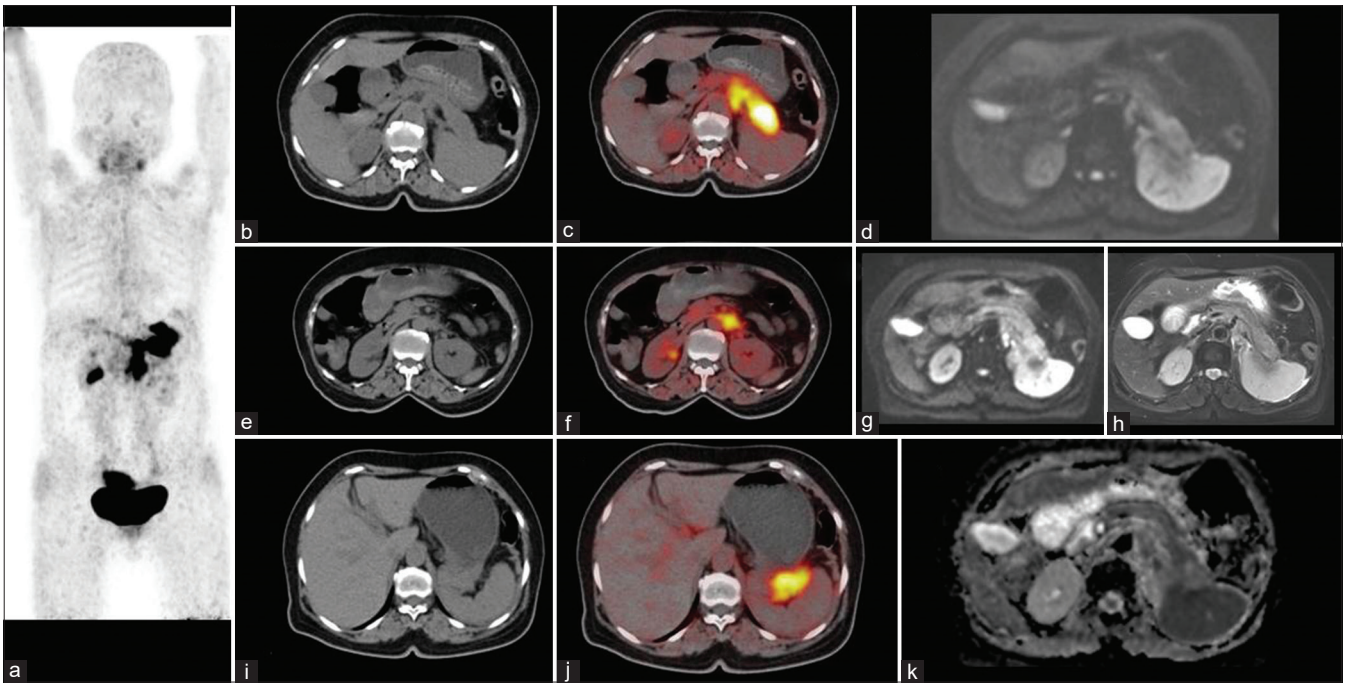


Figure 5: A 65 years old woman presented with pain in abdomen. Sonography showed bulky pancreas. Upper GI endoscopy revealed gastritis.Fibroblast activation protein inhibitor (FAPI) scan showed avid tracer uptake in the bulky tail of pancreas (b,c). MRI shows evidence of restricted diffusion within this region (d, g, h and k). Left para-aortic lymph node also shows avid FAPI uptake (e, f). There was serosal infiltration/ thickening with FAPI uptake in the region of fundus of stomach (i, j). MIP (maximum intensity projection) image did not show involvement of distant organs (a). Histology was adenocarcinoma pancreas

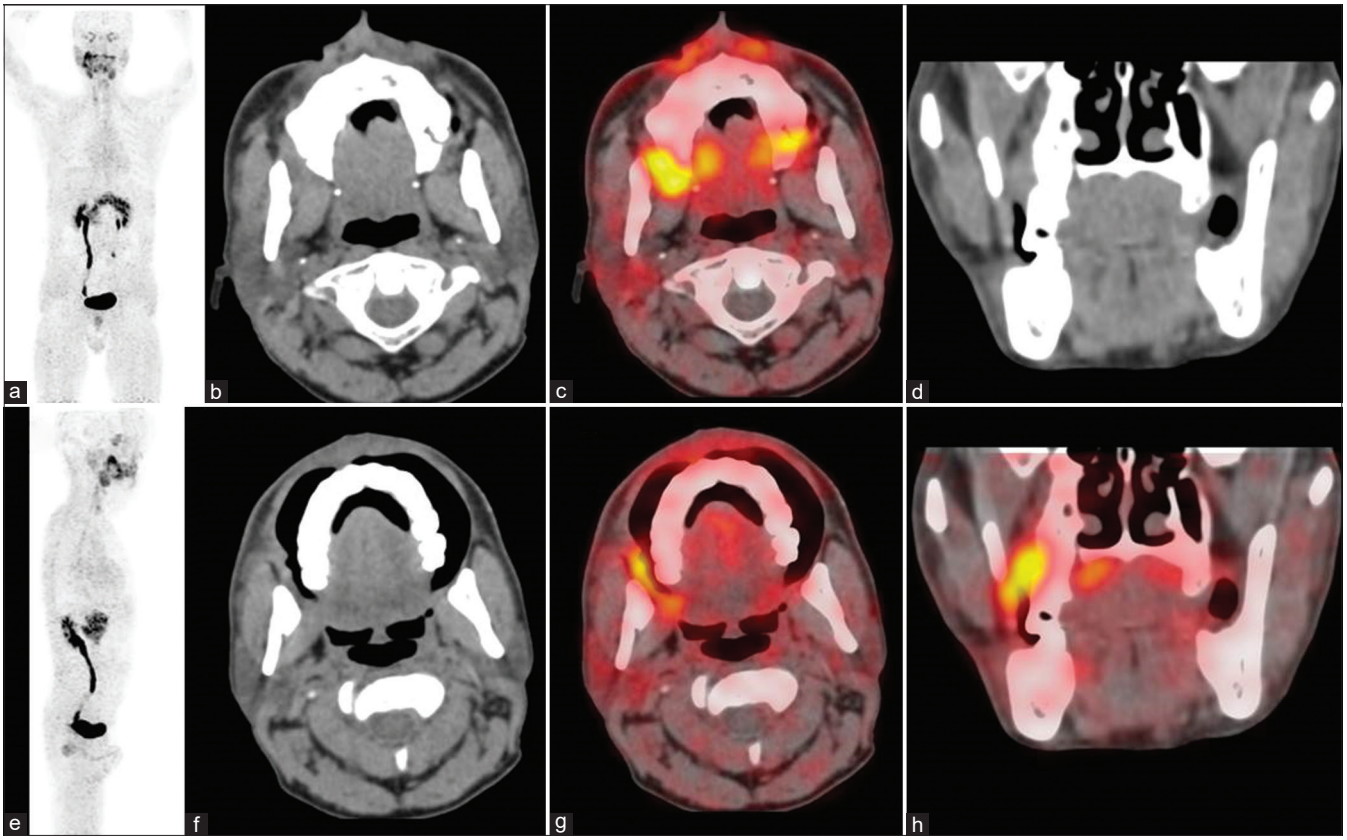


Figure 6: A 43 years male with moderately differentiated squamous cell carcinoma within an ulcero-proliferative growth in buccal mucosa had two cycles of chemotherapy. FAPI scan showed avid tracer uptake in the right retromolar trigone (b and c), extension in the posterior aspect of right buccal mucosa (f and g), as well as involvement of right superior gingivobuccal sulcus (d and h). MIP (maximum intensity projection) images did not show involvement of distant organs. (a and e)

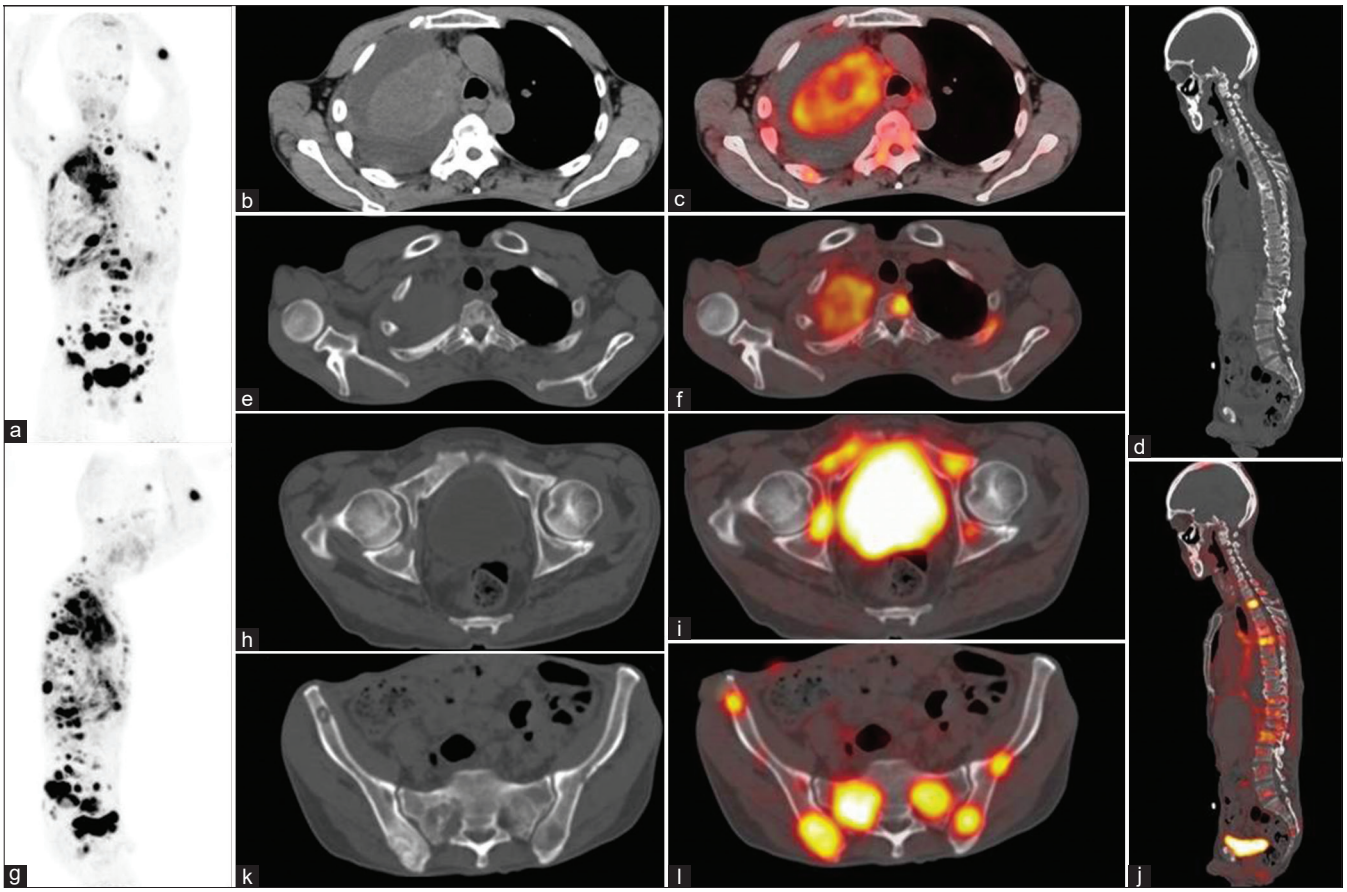


Figure 7: A 45 years old man presented with dyspnoea and dry cough. Plain chest radiograph revealed pleural effusion. Pleural cytology was suggestive of malignant effusion. Fibroblast activation protein inhibitor (FAPI) scan showed avid tracer localization at multiple sites on MIP images (a,g), FAPI avid mass in collapsed right lung upper lobe (b,c), multiple FAPI avid skeletal deposits (d-f,h-l)

fibroblast activation associated with the tumor. Further evaluation, including histology, is essential for the accurate diagnosis and staging.

The incidental liver lesion showed minimal FAPI uptake, primarily at its periphery. MRI findings indicating features of a hydatid cyst are relevant. Hydatid cysts are caused by the parasite Echinococcus granulosus and can occur in various organs, including the liver. Timely assessment and collaboration with other imaging modalities are crucial for accurate diagnosis.

Urothelial Carcinoma and Synonym-transitional Cell Carcinoma

Urothelial carcinoma (UC) arises from the urothelial lining of the urinary tract, including the renal pelvis, ureters, bladder, and urethra; it is the most common type of bladder cancer and can also occur in other parts of the urinary system. UC exhibits variable behavior, ranging from low-grade noninvasive tumors to aggressive invasive carcinomas.

Immunohistochemical studies have revealed a high expression of FAP in UCs. FAP-based imaging with Ga68-FAPI PET/CT shows potential diagnostic efficacy in muscle invasive bladder carcinoma and metastatic UC.

Figure 9 shows FAPI avid soft-tissue lesion in the wall of dilated renal pelvis that was subsequently proven to be UC. A small cohort of eight cases studied by Novruzov *et al*. showed higher tumor to the background ratio in FAPI as compared to FDG.[28]

While Ga-68-FAPI PET/CT provides valuable insights, some lesions, including RCCs, neuroendocrine tumors, pheochromocytomas, and medullary thyroid carcinomas, may show relatively weak FAPI uptake. In such cases, alternative tracers such as F18-dihydroxyphenylalanine or Ga68-Somatostatin analogs can be considered for more specific imaging.[29]

Personalized approaches based on tumor type and clinical context are essential for accurate diagnosis and management.

Liver Disease

While the study primarily focused on cancer detection, it is interesting to note that Ga-68-FAPI PET has shown promise in detecting the early phases of chronic liver disease. Ga68-FAPI PET/CT demonstrated high sensitivity in detecting hepatic malignancies, particularly in poorly differentiated forms with elevated FAP expression. Our

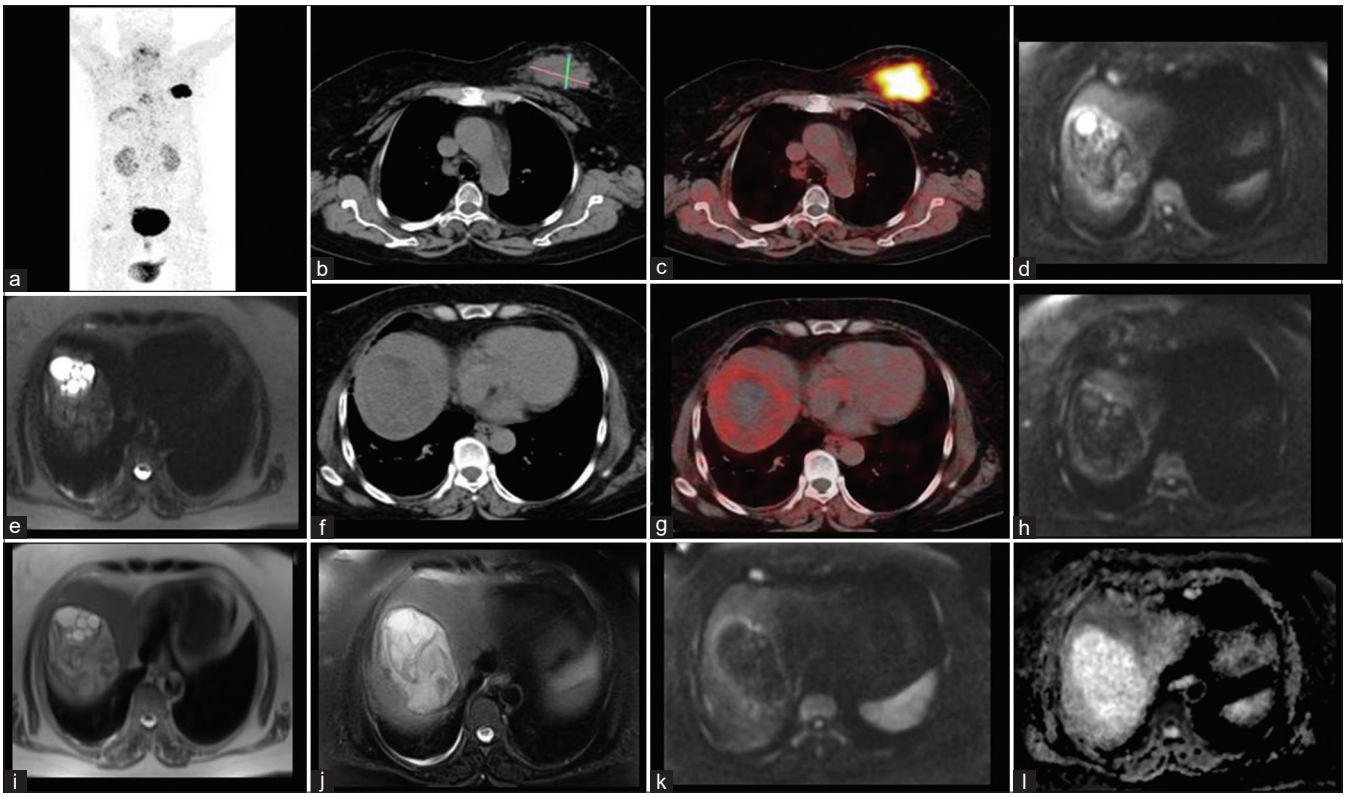


Figure 8: A 61 years old woman had biopsy proven ductal carcinoma of breast. Sonography abdomen showed liver mass suspicious for cystic metastasis or hydatid disease. On arrival her blood sugar was 45mg/ dl. Fibroblast activation protein inhibitor (FAPI) scan showed avid tracer uptake in left breast mass (b and c). Small axillary nodes with FAPI uptake. Liver mass was devoid of FAPI uptake (f and g). Magnetic resonance imaging (MRI) of liver showed cystic lesion with daughter cysts compatible with hydatid disease.(d, e, h-l). Maximum intensity projection (MIP) image showed abnormality localized to the left breast. (a)

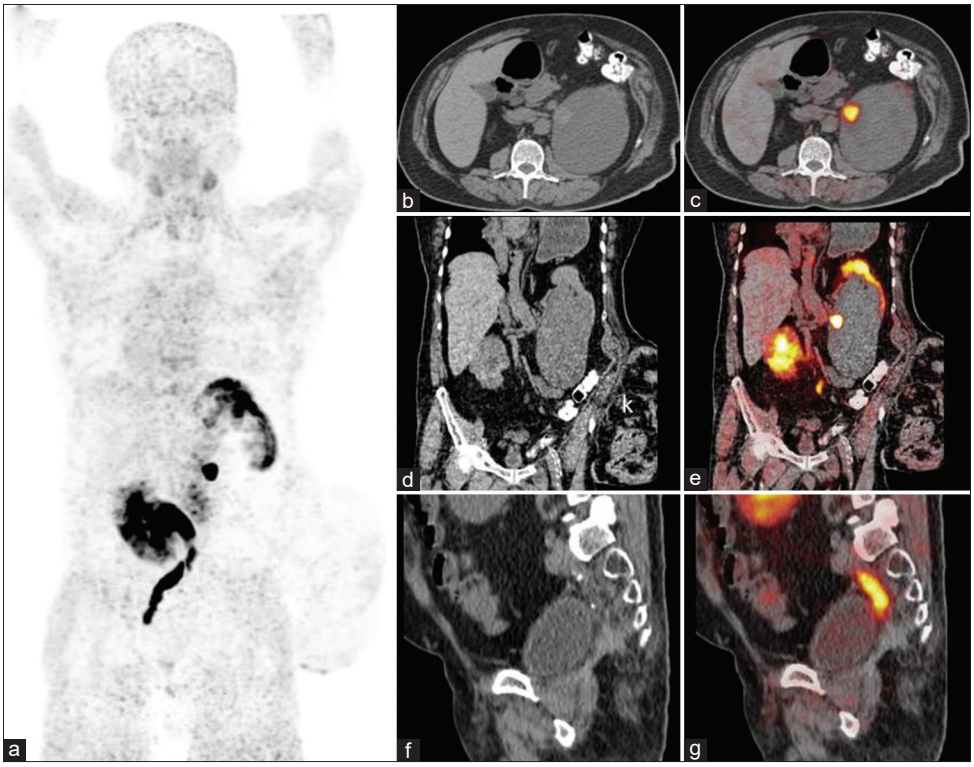


Figure 9: A 64 years old man had computed tomography of kidneys that showed a nodular enhancing lesion in the dilated renal pelvis. Fibroblast activation protein inhibitor (FAPI) scan showed intense tracer localization within the nodular lesion in medial aspect of left renal pelvis (b-e).The anterior and posterior walls of bladder did not show any abnormal focus of tracer uptake. (f, g). Maximum intensity projection (MIP) scan showed physiologic excretion from right kidney with no distant organ involvement. (a). Histology was urothelial carcinoma

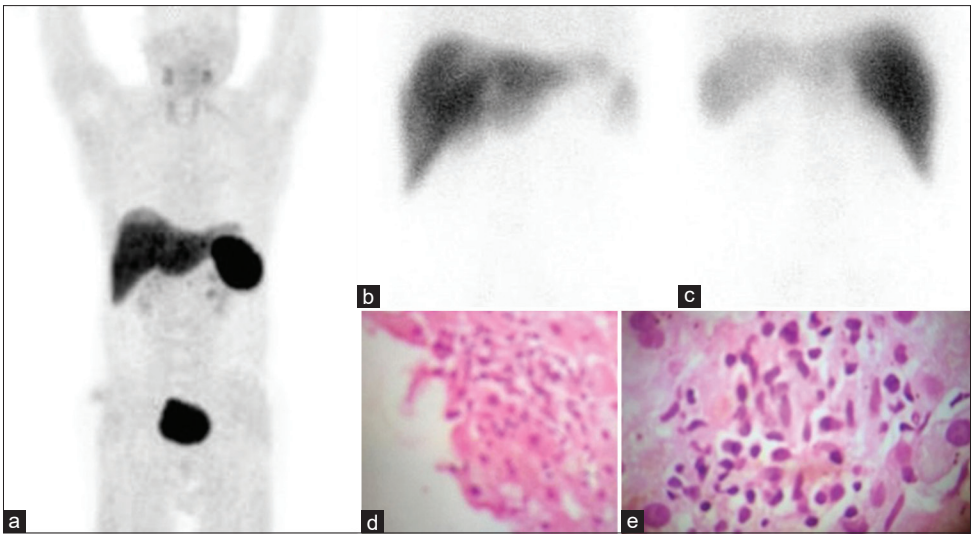


Figure 10: A 52 years old man with epigastric pain and 2 kg weight loss in few weeks had normal clinical examination. His liver functions and sonography of abdomen was normal. The upper gastrointestinal endoscopy showed antral gastritis. Fibroblast activation protein inhibitor (FAPI) scan showed diffuse increased tracer localization in the liver (a). A mesenteric lymph node with minimal FAPI uptake and restricted diffusion on magnetic resonance imaging was indeterminate (not shown here). Colloid liver scan showed inhomogeneous distribution of tracer in liver but no frank features of chronic liver disease (b and c). Biopsy of the liver showed mild peri portal inflammatory changes (d and e)

case [Figure 10] shows diffuse FAPI uptake in the liver parenchyma. A possibility of chronic liver disease was raised despite the colloid liver scan showing no obvious chronic liver disease. Histology showed mild peri-portal fibrosis. Thus, Ga68-FAPI appears to be promising in detecting early phase of chronic liver disease. Benign entities such as hepatic fibrosis, exostosis, and hemorrhoids have been reported to localize FAPI.[30]

In summary, Ga-68-FAPI PET shows promise not only in cancer imaging but also in detecting inflammatory liver disease.

Declaration of patient consent

The authors certify that they have obtained all appropriate patient consent forms. In the form the patients have given their consent for their images and other clinical information to be reported in the journal. The patients understand that their names and initials will not be published and due efforts will be made to conceal their identity, but anonymity cannot be guaranteed.

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Conflicts of interest

There are no conflicts of interest.

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